

EXPERIMENT – 10

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USER INTERFACE AND DESIGN

Data Visualization for Inventory Management System

AIM

To create data visualizations such as pie charts and bar graphs for an inventory management system to represent stock distribution effectively.

OBJECTIVE

- To visualize inventory data using charts.
- To analyze stock distribution across categories.
- To improve data understanding through graphical representation.

CATEGORIES USED

- Electronics
- Clothing
- Home Appliances
- Books
- Toys

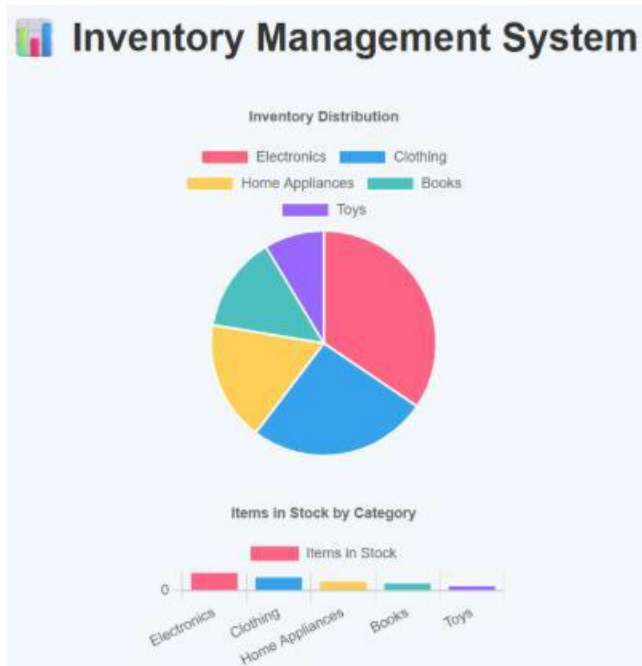
HCI PRINCIPLES USED

- Visibility
- Simplicity
- Data clarity
- Consistency

DESCRIPTION

The experiment includes creating:

- Pie chart for inventory distribution
- Bar graph for category-wise stock analysis



The visualization helps users quickly understand stock availability in different product categories.

PYTHON CODE

```
import matplotlib.pyplot as plt
```

```
# Inventory data
```

```
categories = ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys']
```

```
stock = [200, 150, 120, 100, 80]
```

```
# Create figure
```

```
plt.figure(figsize=(10,5))
```

```
# Pie Chart
```

```
plt.subplot(1,2,1)
plt.pie(stock, labels=categories, autopct='%1.1f%%') plt.title("Inventory
Distribution")
```

```
# Bar Graph plt.subplot(1,2,2)
plt.bar(categories, stock)
plt.title("Items in Stock by Category")
plt.xlabel("Categories") plt.ylabel("Items
in Stock")
```

```
# Display graphs
plt.tight_layout()
plt.show()
```

WORKING PROCESS

1. Define inventory categories and stock values.
2. Create a pie chart to show stock distribution.
3. Create a bar graph for category comparison.
4. Display visualizations using Matplotlib.

RESULT

The pie chart and bar graph successfully represented inventory stock distribution and category-wise stock analysis.